Overview

* MC-equivalent normalization · ≥ observed calls · † first-order norm propagation may be unreliable

Finite coefficient ε^0 · 25-core integration

Massive bubble

B0(16;3,3), μ^2=1 · average work target 30M / sector

Prescription	Sampler	Generation s	Gen. RAM GB	Artifacts MB	Sectors	f64 max / avg µs / sample	Int. RAM GB	Σ f64 / checked µs / sweep	Average work rel. σ(|I|)*	C	max / |I|	Sector calls 600 s · million	Rel. σ(|I|) 600 s*
Fixed	QMC	0.886	0.0189	0.0178	1	0.149 / 0.149	0.297	0.149 / 0.246	1.88e-17	1.27	3.19	344	**6.26e-18**
	MC						0.658		0.000114	1.27	1.8	247	**4e-05**
Dynamical	QMC	0.861	0.0253	0.0246	1	0.291 / 0.291	0.316	0.291 / 0.389	2.96e-16	2.81	7.91	312	**9.88e-17**
	MC						0.659		0.000272	2.81	4.64	247	**9.47e-05**
Sign-aware	QMC	0.865	0.0508	0.0245	1	0.298 / 0.298	0.567	0.298 / 0.393	2.95e-16	2.81	7.91	338	**9.85e-17**
	MC						0.61		0.00027	2.81	4.64	218	**0.0001**
No contour	QMC	0.542	0.0569	0.0675	6	0.0768 / 0.0716	0.377	0.429 / 1	9.41e-17	1.27	1.07	243	**7.18e-17**
	MC						0.429		9.24e-05	1.27	2.74	114	**0.000118**

Massive triangle

C0(0,0,5;1,1,1), μ^2=1 · average work target 80M / sector

Prescription	Sampler	Generation s	Gen. RAM GB	Artifacts MB	Sectors	f64 max / avg µs / sample	Int. RAM GB	Σ f64 / checked µs / sweep	Average work rel. σ(|I|)*	C	max / |I|	Sector calls 600 s · million	Rel. σ(|I|) 600 s*
Fixed	QMC	1.01	0.0508	0.0307	2	0.17 / 0.167	0.323	0.334 / 0.536	2.7e-10	1.35	17.1	333	**2.08e-10**
	MC						0.642		5.71e-05	1.35	4.21	252	**4.54e-05**
Dynamical	QMC	1.01	0.0191	0.0466	2	0.341 / 0.34	0.316	0.681 / 0.887	8.62e-12	1.26	5.66	312	**6.13e-12**
	MC						0.636		5.21e-05	1.26	2.23	252	**4.18e-05**
Sign-aware	QMC	1.01	0.0476	0.0466	2	0.343 / 0.341	0.323	0.682 / 0.896	8.63e-12	1.26	5.66	339	**6.14e-12**
	MC						0.639		5.29e-05	1.26	2.23	252	**4.2e-05**
No contour	QMC	0.692	0.24	0.35	18	0.0974 / 0.0729	0.443	1.31 / 3.09	5.18e-13	1.58	2.16	340	**9.06e-13**
	MC						0.389		3.1e-05	1.58	1.79	61.9	**0.000151**

Massive box

D0(0,0,0,0,5,−1;1,1,1,1,1), μ^2=1 · average work target 70M / sector

Prescription	Sampler	Generation s	Gen. RAM GB	Artifacts MB	Sectors	f64 max / avg µs / sample	Int. RAM GB	Σ f64 / checked µs / sweep	Average work rel. σ(|I|)*	C	max / |I|	Sector calls 600 s · million	Rel. σ(|I|) 600 s*
Fixed	QMC	1.16	0.0633	0.0442	2	0.209 / 0.205	0.326	0.41 / 0.622	0.000112	2.23	216	369	**5.68e-05**
	MC						0.661		0.000183	2.23	26.2	251	**0.000139**
Dynamical	QMC	1.16	0.0254	0.0705	2	0.407 / 0.399	0.323	0.799 / 1.01	1.83e-08	1.63	29.3	343	**1.02e-08**
	MC						0.673		0.000114	1.63	6.45	255	**8.5e-05**
Sign-aware	QMC	1.16	0.0601	0.0705	2	0.407 / 0.4	0.326	0.801 / 1.02	1.83e-08	1.63	29.3	334	**1.02e-08**
	MC						0.638		0.000115	1.63	6.31	255	**8.48e-05**
No contour	QMC	1.8	0.0949	2.36	54	0.161 / 0.114	0.693	6.13 / 12.5	2.27e-10	4.79	16.5	305	**5.33e-10**
	MC						0.506		8.41e-05	4.79	4.99	29.4	**0.000973**

Overview

* MC-equivalent normalization · ≥ observed calls · † first-order norm propagation may be unreliable

Finite coefficient ε^0 · 25-core integration

Massless sunrise

$p^2=1$; all three masses zero · average work target 60M / sector

Prescription	Sampler	Generation s	Gen. RAM GB	Artifacts MB	Sectors	f64 max / avg µs / sample	Int. RAM GB	Σ f64 / checked µs / sweep	Average work rel. σ(|I|)*	C	max / |I|	Sector calls 600 s · million	Rel. σ(|I|) 600 s*
Fixed	QMC	1.43	0.0886	0.0238	1	0.16 / 0.16	0.31	0.16 / 0.26	4.01e-13	1.01	5.66	153	**2.7e-13**
	MC						0.358		2.58e-05	1.01	3.21	120	**1.84e-05**
Dynamical	QMC	1.49	0.0887	0.027	1	0.157 / 0.157	0.319	0.157 / 0.257	4.04e-13	1.01	5.66	149	**2.7e-13**
	MC						0.362		2.6e-05	1.01	3.46	126	**1.81e-05**
Sign-aware	QMC	1.45	0.0728	0.0242	1	0.162 / 0.162	0.313	0.162 / 0.263	4.07e-13	1.01	5.66	151	**2.71e-13**
	MC						0.37		2.58e-05	1.01	3.07	130	**1.77e-05**
No contour	QMC	0.539	0.0254	0.104	6	0.187 / 0.179	0.501	1.08 / 1.66	1.67e-13	1.01	1.19	306	**2.23e-13**
	MC						0.4		1.05e-05	1.01	0.426	106	**1.94e-05**

Massive sunrise

m1=m2=m3=1, s=16; native measure1 · average work target 0.323M / sector

Prescription	Sampler	Generation s	Gen. RAM GB	Artifacts MB	Sectors	f64 max / avg µs / sample	Int. RAM GB	Σ f64 / checked µs / sweep	Average work rel. σ(|I|)*	C	max / |I|	Sector calls 600 s · million	Rel. σ(|I|) 600 s*
Fixed	QMC	1.61	0.0444	0.0365	1	0.652 / 0.652	0.485	0.652 / 1.74	1.07e-11	1.25	5.88	232	**5.27e-13**
	MC						0.635		0.00097	1.25	9.83	213	**3.78e-05**
Dynamical	QMC	1.44	0.0381	0.0801	1	3.62 / 3.62	0.568	3.62 / 7.09	8.75e-10	3.84	23.7	160	**4.29e-11**
	MC						0.608		0.00545	3.84	49.5	207	**0.000216**
Sign-aware	QMC	1.44	0.0413	0.0859	1	2.47 / 2.47	0.568	2.47 / 5.12	4.38e-09	3.87	24.1	154	**2.15e-10**
	MC						0.628		0.00548	3.87	45.8	202	**0.00022**
No contour	QMC	6.7	0.215	2.3	62	2.79 / 0.764	0.497	47.4 / 2.37e+04	3.65e-11	1.54	4.15	25.9	**3.95e-11**
	MC						0.561		0.000301	1.54	1.66	14.8	**0.000351**

Overview

* MC-equivalent normalization · ≥ observed calls · † first-order norm propagation may be unreliable

Finite coefficient ε^0 · 25-core integration

Massive-threshold kite

p^2=16; five internal masses equal to 1 · average work target 40M / sector

Prescription	Sampler	Generation s	Gen. RAM GB	Artifacts MB	Sectors	f64 max / avg µs / sample	Int. RAM GB	Σ f64 / checked µs / sweep	Average work rel. σ(|I|)*	C	max / |I|	Sector calls 600 s · million	Rel. σ(|I|) 600 s*
Fixed	QMC	2.98	0.238	0.59	4	0.445 / 0.42	0.326	1.68 / 2.14	2.71e-05	1.86	17.2	377	**1.82e-05**
	MC						0.599		0.000125	1.86	11.8	205	**0.000111**
Dynamical	QMC	3.93	0.275	4.05	4	4.13 / 3.1	0.405	12.4 / 12.9	0.000164	2.5	57.8	353	**0.000147**
	MC						0.672		0.00065	2.5	3.03e+03	209	**0.000408**
Sign-aware	QMC	5.18	0.319	7.01	4	4.76 / 3.68	0.483	14.7 / 15.2	0.000185	2.6	59.8	351	**0.000172**
	MC						0.641		0.000518	2.59	1.23e+04	207	**0.000853**
No contour	QMC	Not completed	—	—	—	—	—	—	—	—	—	—	—
	MC						—		—	—	—	—	—

Overview

* MC-equivalent normalization · ≥ observed calls · † first-order norm propagation may be unreliable

Scalar D05 double box

s=1000000, t=−97077, mH=125, mt=345/2 · average work target 4M / sector

Prescription	Sampler	Generation s	Gen. RAM GB	Artifacts MB	Sectors	f64 max / avg µs / sample	Int. RAM GB	Σ f64 / checked µs / sweep	Average work rel. σ(|I|)*	C	max / |I|	Sector calls 600 s · million	Rel. σ(|I|) 600 s*
Fixed	**QMC**	9.77	0.561	28.8	18	0.738 / 0.724	0.475	13 / 15.1	0.0259	9.95	7.02e+03	363	**0.0102**
	MC						0.653		0.00256	9.98	740	85.3	**0.00246**
Dynamical	**QMC**	19.2	1.46	77.8	18	5.77 / 4.68	0.64	84.3 / 87	0.306†	19.8	1.43e+07	342	**0.169†**
	MC						0.837		0.286†	14.3	1.09e+06	67.1	**0.167†**
Dynamic tuned	**QMC**	19.2	1.46	77.8	18	6.2 / 5.33	0.956	95.9 / 98.5	0.0118	9.37	1.75e+03	289	**0.00665**
	MC					5.14 / 4.4	0.862	79.2 / 81.8	0.00271	10	692	77.3	**0.00262**
Sign-aware	**QMC**	34.1	2.03	88.6	18	6.74 / 5.47	1.13	98.5 / 101	0.328†	24.4	4.04e+06	326	**0.197†**
	MC						0.965		0.626†	35.2	5.04e+06	82.5	**0.586†**
Sign-aware tuned	**QMC**	34.1	2.03	88.6	18	7.36 / 6.16	1.08	111 / 114	0.0125	10.7	1.98e+03	284	**0.00755**
	MC					6.16 / 5.23	0.997	94.2 / 96.7	0.00248	10	546	82.5	**0.00233**

Overview

* MC-equivalent normalization · ≥ observed calls · † first-order norm propagation may be unreliable

Physical D05 double box

s=1000000, t=−97077, mH=125, mt=345/2; incoming ++ · average work target 6M / sector

Prescription	Sampler	Generation s	Gen. RAM GB	Artifacts MB	Sectors	f64 max / avg µs / sample	Int. RAM GB	Σ f64 / checked µs / sweep	Average work rel. σ(|I|)*	C	max / |I|	Sector calls 600 s · million	Rel. σ(|I|) 600 s*
Fixed	**QMC**	253	8.99	38.1	30	3.22 / 3.01	0.476	90.4 / 94.5	0.0311	23.1	1.04e+04	344	**0.0198**
	MC						0.755		0.00329	23.2	1.03e+03	44.6	**0.00594**
Dynamical	**QMC**	1.8e+03	34.8	88.6	30	8.08 / 6.93	0.54	208 / 213	0.336†	44.5	2.81e+08	328	**0.457†**
	MC						1.09		0.308†	200	6.95e+05	44.2	**1.95†**
Dynamic tuned	**QMC**	1.8e+03	34.8	88.6	30	7.52 / 6.71	0.775	201 / 206	0.0272	22.6	9.79e+03	297	**0.0209**
	MC					7.52 / 6.71	1.14	201 / 206	0.00324	23.1	1.43e+03	43.6	**0.00662**
Sign-aware	**QMC**	1.54e+03	36.7	99.7	30	9.11 / 7.72	0.805	232 / 237	0.924†	98.8	8.71e+07	332	**0.708†**
	MC						1.2		0.333†	61.6	7.27e+05	43.6	**0.68†**
Sign-aware tuned	**QMC**	1.54e+03	36.7	99.7	30	8.98 / 7.59	0.807	228 / 232	0.0287	22.8	9.69e+03	323	**0.0214**
	MC					8.98 / 7.59	1.2	228 / 232	0.00313	23	1.12e+03	42.4	**0.00648**

Threshold regularisation in FastSecDec

Methods, measurements and limits

Campaign evidence through 12 October 2026

The opening tables are the primary benchmark comparison: the finite Laurent coefficient of eight topologies, with QMC and discrete MC shown together. The chapters that follow explain how to read those measurements, what the algorithms compute, and which conclusions the available evidence supports. Complete integral results, local preparation controls and future research proposals are treated separately throughout.

Reading guide. Start with Chapter 1 for column definitions, inputs and reproduction. Chapter 2 defines fixed, dynamic and no-contour prescriptions; Chapter 3 interprets their measured accuracy and compatibility. Chapter 4 explains the principal performance choices. Chapter 5 records unfinished algebraic boundaries, and Chapter 6 develops the critical-locus and region-analysis research programme.

Contents

1 Campaign design and reading guide 7
1.1 What the columns measure 7
1.2 Uncertainty and display normalization 7
1.3 Execution, clocks and numerical policy 8
1.4 Topologies and runtime settings 8
1.5 Reproduction and evidence boundaries 9
2 Threshold-regularisation methods 10
2.1 Common geometry and fixed strength 10
2.2 Branches and checking policy 10
2.3 Dynamic strength: a smooth sufficient causal radius 11
2.4 Full local-strength derivatives 11
2.5 Sign-aware refinement of the dynamic bound 12
2.6 Sampling, diagnostics and interpretation 12
2.7 Real-domain threshold decomposition 13
3 Benchmark findings and scientific checks 14
3.1 Scalar topologies: accuracy versus representation cost 14
3.2 Precision rescue changes the effective cost 14
3.3 Double boxes: tuning restores fixed-like behavior 15
3.4 Compatibility and independent references 16
4 Implementation and performance 17
4.1 Preserve structure through generation and restoration 17
4.2 Symbolic subtraction with two Jacobian implementations 17
4.3 Selective residency and native evaluator loading 18
4.4 Deformation clearance and Jacobian amplification 18
4.5 Statistical ownership and focused reproduction 18
5 Support limits and algebraic boundaries 19
5.1 Double box: complete geometry, incomplete density resolution 19
5.2 What the local certificates establish 20
5.3 Massive kite: a boundary is not a complete interval 20
6 Critical loci and future region analysis 21
6.1 Stationary zeros, causal pinches and present diagnostics 21
6.2 Equality strata: how exact algebra and CAD can cooperate 21
6.3 From a certified pinch to local subtraction 22
6.4 Prospects for complete expansion by regions 24

Campaign design and reading guide

The overview contains **66 measured sampler rows across eight topologies**, plus two unavailable massive-kite no-contour slots. It compares complete integrals through I_0, retaining lower Laurent orders and covariance. A dash is missing evidence, never zero. No-contour double-box integration was removed from scope; its partial preparation appears in Chapter 5.

1.1 What the columns measure

Generation is wall time from launching generation to durable artifact publication, excluding the software build. **Artifacts** counts the manifest and referenced indexed data once. **Sectors** counts executable stochastic kernels; geometry cells, endpoint charts and exact contributions are distinct objects. This distinction is essential when reading partial GCAD results.

Generation RAM and **integration RAM** are sampled aggregate resident memory of the owned process tree, in decimal GB. Shared mappings can be counted more than once and short peaks can be missed. These measurements are neither unique memory nor an exact high-water mark.

f64 max / avg is the maximum and arithmetic mean of per-sector mean raw kernel costs, not the slowest individual sample. **Σ f64 / checked** sums those means for one sweep over all sectors; the checked cost retains ordinary stability handling and precision rescue. It is not a parallel wall-time prediction. A sparse profile need not reproduce a long integration's rescue frequency; profiling settings are recorded below.

Sector calls counts logical sector-integrand evaluations during the timed invocation, including reported pilot and discarded work. Replaying a point at higher precision is a backend call, not an additional sampling point. Unprefixed totals require complete native deadline/drain accounting; $\geq$ marks an older observed prefix. Accepted statistical work is recorded separately and exactly.

The cancellation diagnostic is

$$C_{\mathrm{abs}} = \frac{\sum_s \int |w_s(u)|\,\mathrm{d}\mu_s(u) + |I_{\mathrm{exact}}|}{|I_0|},$$

where w_s is the final complex finite-coefficient weight, including its continuous importance and coordinate factors. The displayed column **C** uses committed production estimates of these absolute-weight integrals. It is not the sum of absolute values of the sector integrals. The combined exact offset enters once. Discrete MC accounts for sector-selection probabilities in this integral estimate. **max / |I|** instead uses the largest observed sector-local complex weight: it removes the discrete selection factor while keeping continuous importance corrections. This proposal-dependent sample maximum is not a supremum or a tail bound.

1.2 Uncertainty and display normalization

For $I_0 = a + ib$ and $M = |I_0|$, the final column reports the relative standard error of the complex **magnitude**. With covariance Σ of the estimated real and imaginary means,

$$\sigma_M^2 = \frac{a^2\Sigma_{RR} + 2ab\Sigma_{RI} + b^2\Sigma_{II}}{M^2}, \quad \text{relative SE} = \frac{\sigma_M}{M}.$$

It is undefined at $M = 0$. A dagger warns that the joint complex standard error $\frac{\sqrt{\mathrm{tr}\,\Sigma}}{M}$ is at least 0.1, where first-order norm propagation may be unreliable. This is a warning convention, not a certified boundary. Radial uncertainty does not measure phase accuracy. Full complex covariance and joint SE remain the basis of compatibility checks; discrete-MC covariance is the native global-batch covariance, not a sum of sector marginals.

The two relative-error columns apply the requested **MC-equivalent display heuristic**: multiply the measured relative SE by $\sqrt{\frac{W}{W_{\mathrm{target}}}}$ for accepted work, or by $\sqrt{\frac{t}{600}}$ for native elapsed seconds. These are not measured QMC convergence laws and never alter estimates or covariance. Average work is total accepted calls divided by stochastic sector count, not equal allocation; targets appear in topology headers. Large corrections require caution. The last, near-ten-minute column is the primary metric.

1.3 Execution, clocks and numerical policy

Each accepted timed run lasts 480–750 native seconds and comparisons within a topology require a longest/shortest ratio at most 1.25. Instrumented runs stop new reservations at 600 seconds, drain complete in-flight work and save the native result. A budget stop is not an accuracy stop. External kills do not qualify. Earlier graceful-cancellation observations retain their prefix qualification rather than being relabelled exact totals.

The integration clock starts after ordinary coordinator loading. Serial resident loading and later reloads occur inside that clock. Process wall, loading, drain and publication receipts are kept separately; their clocks must not be added as though they were disjoint. Integration RAM comes from the timed arm. Independent throughput calibrations select work allocations; confirmation streams are disjoint from calibration and other arms. Different designs, lattice epochs, seeds and covariance matrices are never pooled to construct the displayed observations.

The secondary work display may retain a clean capped MC estimate whose planned allocation is incomplete, provided committed production batches give full coverage and available covariance under one fixed design. It uses the actual count and remains explicitly incomplete requested work. A completed continuation replaces its overlapping prefix. Some newer extensions use the timed run itself for the secondary display; their receipts identify that choice. No such correction creates a matched-work experiment.

Generation uses serial sector residency with 25 workers, except for the physical-numerator double box's six-worker memory allowance. Integration uses 25 workers. QMC has serial residency with a one-second minimum stay; MC uses the existing **ordinary discrete-MC scheduler**, the explicit exception. The owned campaign cohort has a 100 GB memory allowance. Runs may overlap; no quiet-machine condition is imposed. Preparation time limits are not integration deadlines or completed-generation measurements.

Optional causal validation is off in every overview timing, including profiling and throughput calibration. The collector rejects a causal pilot or nonzero production causal-check count. Mandatory root termination, nonfinite handling, stability checks and precision rescue remain active. Separate finite causal pilots support admission of the retained double-box settings, but are neither timed here nor global certificates. New generation uses Horner ten and SymJIT O2. Historical Horner-zero studies elsewhere in the repository are not substituted for this campaign.

1.4 Topologies and runtime settings

Input	Definition
Bubble	$B_0(16;3,3)$, $\mu^2=1$; arguments after the semicolon are squared masses.
Triangle	$C_0(0,0,5;1,1,1)$, $\mu^2=1$.
Box	$D_0(0,0,0,0,5,-1;1,1,1,1)$, $\mu^2=1$.
Massless sunrise	$p^2=1$, three massless lines; a branch/endpoint control without an interior threshold zero.
Massive sunrise	$p^2=16$, three masses equal to one, native measure one.
Massive kite	$p^2=16$, all five masses equal to one. The earlier mixed-mass small-momentum kite is a different input.
D05 double boxes	$s=10^6$, $t=-97077$, $u=-871673$, $m_H=125$, $m_t=\frac{345}{2}$. Scalar numerator one; physical input uses the original contracted numerator.

All inputs use $D=4-2\varepsilon$. Native cards retain exact masses, invariants, normalization, graphs and symbolic endpoint subtraction. The physical D05 case is an individual s-channel contribution with incoming $++$ helicities, unnormalized external color contraction δ_{ab} and loop measure $\prod_l \mathrm{d}^D \frac{k_l}{i\pi^{\frac{D}{2}}}$, with multiplier one. It is not a complete diagram sum or a spin/color-averaged matrix element. HEPKit supplies the graph, contracted numerator, wavefunctions and kinematics.

Fixed strength is $\lambda=0.1$ for the smaller scalar cases and 10^{-6} for D05. The D05 trial 0.1 failed a separate causal admission check and is not a benchmark result. Dynamic baselines use $S=0.8$, $L=R=1$. Tuned D05 rows use $S=0.8$, $R=10^6$, with $L=1.25\times10^{-5}$ for scalar QMC and 1.25×10^{-6} for scalar MC and both physical samplers. The settings are frozen before confirmation; their interpretation follows in the results chapter. Binding these runtime caps does not regenerate an artifact.

1.5 Reproduction and evidence boundaries

The current measurement source is `docs/benchmark-overview-results.json`. It preserves invocation arguments, binary and artifact identities, complete Laurent vectors, covariance, work ledgers, diagnostics and data-source associations. The exact collection contract and commands are in `docs/benchmark-overview-protocol.md`; native input cards are in `examples/no_deformation/benchmarks/overview/` and the explicitly associated control directories. The generation/profile receipts distinguish symbolic and contour-only dual Jacobians while keeping endpoint subtraction symbolic. Candidate selection precedes confirmation and uses measured full-sweep cost.

The native profiler uses deterministic coordinates, three repetitions and zero warmup: 4096 coordinates for the original smaller scalars, 1024 for the double boxes and later bubble/massive-sunrise measurements. Actual kernel calls and coordinate counts are recorded separately.

The baseline campaign pins symGCAD `644a560e69a064f811fa93aff33a247ea7fcdfca`. Original instrumented timings use executable identity `cf9eb450`; the admitted triangle, massless-sunrise and box routes use `35c92289`, `3cf81e4b` and `d94dd90d`, respectively. Massive-sunrise and final double-box tuning receipts identify frozen source `6647950`. These are separate recorded builds, not a claim that all rows came from one binary. Later algorithmic discussion must not retroactively attribute changes to those measurements.

Evidence	Compact record in `docs/`
Main comparison	`benchmark-overview-results.json`; `benchmark-overview-protocol.md`
Independent references	`benchmark-compatibility-references.json`; `benchmark-compatibility-results.json`
Massive sunrise	`no-contour-massive-sunrise-overview.json`; `massive-sunrise-contour-overview.json`
Sign-aware confirmations	`bubble-sign-aware-overview.json`; `scalar-double-box-sign-aware-overview.json`; `physical-double-box-sign-aware-overview.json`
Runtime-cap tuning	`double-box-dynamic-tuning-overview.json`; `double-box-dynamic-work-overview.json`
Scope and acceptance	`reviews/benchmark-final-scope.md`; `reviews/no-contour-final-scope.md`

Historical five-minute/50-core runs, Horner-zero studies, early refusal receipts and one-worker correctness probes remain available in their compact source records. They are not additional rows of the current comparison and are not used to fill missing measurements. Older reports of unsupported triangle/box routes are superseded by complete artifacts and fresh confirmations; the massive-kite interval/whole-atlas gap remains real.

The campaign is complete under the final narrowed scope. This does not mean that the general algebraic resolver is complete, that fixed deformation is globally certified by finite pilots, or that a small stochastic SE certifies floating-point accuracy. The benchmark findings distinguish these limits from measured efficiency and reference agreement.

Threshold-regularisation methods

The benchmark compares two ways to represent a causal integral: deform its parameter contour, or partition the real domain and continue the resulting endpoint integrals analytically. Both preserve the original numerator, normalization and complete Laurent vector. The second route additionally requires endpoint certificates; a successful sign decomposition alone is insufficient. This chapter defines the constructions behind the overview.

2.1 Common geometry and fixed strength

After sector mapping, separate the real endpoint monomials from the smooth density. Let $F(x)$ denote the designated residual causal polynomial and let $U(x)$ denote each retained positive residual factor. The definition of F includes its physical scale: rescaling F also rescales this deformation. For a sampled sector with real masses, real kinematics and $x \in [0,1]^n$ ($n \geq 1$), define

$$w_i = x_i(1-x_i), \quad v_i = w_i \partial_i F, \quad z_{i(x)} = x_i - i\lambda(x) v_{i(x)}.$$

Exact zero-dimensional contributions bypass sampling. The coordinate faces remain fixed in their normal direction. With $A = \sum_i w_i (\partial_i F)^2 \geq 0$, the ray obeys

$$\operatorname{Im} F(x - itv) = -tA + O(t^3).$$

Thus a sufficiently small positive strength moves the causal denominator to the lower half-plane. The scalar λ is a strength, not the norm of the displacement; no undocumented normalization by F or its gradient is used. Since cube coordinates are dimensionless, $[\lambda] = [F]^{-1}$; the same units apply to L. The displacement limit R is in parameter-space units.

Fixed mode takes a finite runtime value $\lambda(x) = \lambda_0 > 0$. Its Jacobian is

$$J_{ij} = \delta_{ij} - i\lambda_0 \partial_j v_i, \quad \partial_j v_i = \delta_{ij}(1-2x_i)\partial_i F + w_i \partial_j \partial_i F.$$

The complex, oriented determinant $\det J$ multiplies the density. A zero determinant alone is not a pole crossing. Neither its absolute value nor a frozen-gradient approximation gives the same contour integral.

For endpoint exponents $a_i + b_i \varepsilon$, the real endpoint monomials are retained and the smooth density receives

$$\det J \prod_i \rho_i^{a_i + b_i \varepsilon}, \quad \rho_i = \frac{z_i}{x_i} = 1 - i\lambda(x)(1-x_i)\partial_i F.$$

The last expression defines ρ_i even at $x_i = 0$. Deform the numerator and residual factors as well. Apply this pullback **before** Taylor/IBP subtraction and Laurent construction. Boundary terms restrict the same full-sector map; they do not construct a new, lower-dimensional deformation.

2.2 Branches and checking policy

Continue each factor separately. Endpoint ratios have real part one; positive residuals must keep the appropriate branch, and the causal logarithm has the lower-lip value $\log(F - i0) = \log(|F|) - i\pi$ when $F < 0$. Combining these continued logarithms into the principal logarithm of their product can change the sheet. A stationary point with $F \neq 0$ is allowed. A surviving singularity at $F = 0$ and $A = 0$ requires an unresolved-deformation diagnostic, not a fabricated zero; a natively cancelled denominator or finite smooth density remains finite.

Optional validation has three policies: **always** checks pilot and production, **pilot** checks separate pilot work and removes optional production checks, and **off**, now the default, removes both. Solver termination, precision rescue and nonfinite handling remain active. The overview's sampling and convergence timings use **off**; separate validation probes are excluded. Fixed-mode failures identify the factor and point; they never silently shrink the requested strength. Finite pilots are safeguards, not global certificates. The gradient-map convention and subtraction ordering follow the established sector-decomposition approach; see pySecDec.

2.3 Dynamic strength: a smooth sufficient causal radius

Set $\lambda(x) = Sr(x)$ with $0 < S < 1$. The radius bounds the **whole homotopy** $x - itv(x)$ for $0 < t \le Sr(x)$. It is not the distance to the nearest complex root: a positive real ray root need not exist, and a root-modulus prescription can vanish at the very physical pole one must avoid.

In the auxiliary ray expansion, hold $v(x)$ fixed. For odd $k \ge 3$, define

$$T_k = D^k F(x)[v^{k-2}, \cdot, \cdot], \quad B_k = \frac{\sum_{i,j} w_i w_j (T_k)_{ij}^2}{(k!)^2}.$$

For every positive residual U, write the real even-ray expansion

$$\operatorname{Re} U(x - itv) = U(x) + \sum_{k \ge 2,\ \text{even}} d_{k(x)} t^k, \quad C_k = \frac{d_k^2}{U(x)^2}.$$

Native polynomial positivity admission supplies a strictly positive residual U on the closed cube; an arbitrary positive floor is not substituted. Let m_F and m_U count the structurally retained odd and even orders. With positive cap L and displacement scale R, solve

$$H(x, r) = \frac{r^2}{L^2} + r^2 \frac{\|v\|^2}{R^2} + m_F \sum_{k \ge 3,\ \text{odd}} B_k r^{2k-2} + \sum_U m_U \sum_{k \ge 2,\ \text{even}} C_k r^{2k} = 1.$$

All coefficients are nonnegative. The cap term makes the positive root unique and simple, with $r \le L$ and $rH_r \ge 2$ at the root. Both constraints are smooth; there is no hard minimum whose derivative would jump.

Why the bound is causal. Put $W = \operatorname{diag}(w_i)$ and $q = \sqrt{W}\operatorname{grad} F$, so $A = q^T q$ and $v = \sqrt{W} q$. The higher odd ray term satisfies

$$\left| D^k \frac{F[v^k]}{k!} \right| \le A \sqrt{B_k}, \quad \sum_k \sqrt{B_k} t^{k-1} \le \sqrt{m_F \sum_k B_k t^{2k-2}}.$$

Consequently higher imaginary terms cannot cancel the leading $-tA$ along the admitted homotopy when $A > 0$. The even-factor bound similarly keeps $\operatorname{Re} U(z) > 0$. The proof does not divide by A, so small gradients do not create an artificial singular bound. At $A = 0$, the stationary-point rules still apply.

2.4 Full local-strength derivatives

Dynamic strength changes the Jacobian:

$$J_{ij} = \delta_{ij} - i\big(\lambda \partial_j v_i + v_i \partial_j \lambda\big).$$

Implicit differentiation of $H(x, r(x)) = 1$ gives

$$\partial_j r = -\frac{\partial_j H}{\partial_r H}, \quad \partial_j \lambda = S \partial_j r.$$

Here $\partial_j H$ holds r fixed but differentiates all physical coefficient dependence, including v and the smooth constraints. Symbolic endpoint reduction differentiates the complete deformed density, including this identity and the determinant. Solver iteration choices are never differentiated. Every face retains the full-sector dimension, structural counts and cap parameters. Omitting the rank-one term $-iv(\operatorname{grad} \lambda)^T$ changes the integral.

The polynomial construction is an independently checked baseline. Its exact-arithmetic proof is distinct from numerical validation of a rounded implementation. Disabling optional checks removes their runtime cost, not solver failures or nonfinite-value diagnostics.

2.5 Sign-aware refinement of the dynamic bound

The baseline bounds beneficial as well as harmful higher terms. A smooth, sign-sensitive alternative bounds terms that can oppose causality. Define the real symmetric matrix

$$R_k = \frac{(-1)^{\frac{k+1}{2}} L^{k-1}}{k!} \sqrt{W} T_k \sqrt{W}.$$

It need not be built with explicit square roots of W. Weighted trace and squared-norm identities provide the native polynomial quantities

$$a_k = \frac{\operatorname{tr}(R_k)}{n}, \quad b_k^2 = \frac{n-1}{n}\left(\operatorname{tr}(R_k^2) - \frac{\operatorname{tr}(R_k)^2}{n}\right).$$

For a symmetric matrix its largest eigenvalue is at most $a_k + \sqrt{b_k^2}$. Smooth both the spectral bound and its positive part with a fixed dimensionless $\delta = 10^{-3}$:

$$t_k = a_k + \sqrt{b_k^2 + \delta^2}, \quad p_{\delta(t)} = \frac{t + \sqrt{t^2 + \delta^2}}{2}, \quad \mu_k = p_{\delta(t_k)}.$$

For negative t, evaluate the exactly equivalent expression

$$p_{\delta(t)} = \frac{\delta^2}{2\left(\sqrt{t^2 + \delta^2} - t\right)}$$

to avoid cancellation. For each even U coefficient, use

$$\nu_{U,k} = p_{\delta\left(-d_k \frac{L^k}{U}\right)}.$$

With $u = \frac{r}{L}$, form

$$E_{F(u)} = \sum_{k \geq 3,\ \text{odd}} \mu_k u^{k-1}, \quad E_{U(u)} = \sum_{k \geq 2,\ \text{even}} \nu_{U,k} u^k.$$

Then solve the unique positive root

$$u^2 + \frac{L^2 u^2 \|v\|^2}{R^2} + E_{F(u)}^2 + \sum_U E_{U(u)}^2 = 1, \quad \lambda = SLu.$$

The nonnegative envelope coefficients retain monotonicity and smoothness. Their derivatives, including smoothing, enter the same implicit-jet machinery. A larger permitted displacement is a diagnostic; it does not establish lower variance or lower time to precision. The baseline and sign-aware construction must be compared with their full evaluation costs.

2.6 Sampling, diagnostics and interpretation

Korobov remains an outer real map: $y = T(q)$ followed by $z = z(y)$. Its total Jacobian is $\det J_{z(y)} \prod_i T_{i'}(q_i)$, with each factor applied once. No inverse Korobov derivative is introduced at an endpoint. Contour evaluation is deterministic and consumes no production random streams.

QMC uncertainties come from complete independently randomized lattices; democratic QMC retains covariance between sectors sharing a shift. Discrete Havana MC retains its frozen production proposal and uses complete batches. Both accumulate the full Laurent vector and its covariance. A partial allocation, a missing sector or an unvisited zero is not exact evidence.

The smooth sufficient-radius bounds above are FastSecDec's parameter-space construction. They are not attributed to pySecDec or to the LTD algorithm. Related work includes Winterhalder et al., *Targeting Multi-Loop Integrals with Neural Networks* on optimized contour choices, and Capatti et al., *Numerical Loop-Tree Duality: contour deformation and subtraction* on causal contour construction and physical multiloop tests. Their timings are not directly comparable FastSecDec speed measurements.

2.7 Real-domain threshold decomposition

The no-contour route uses the verified generic cylindrical algebraic decomposition of the causal geometry, following the developments in the 2025 no-contour construction and its 2026 GCAD extension. FastSecDec calls symGCAD through its native Rust interface and independently replays the returned proof. The physical density is transported to each admitted cell; no denominator sign is silently removed.

For a triangular cell $a_{i(x_{<i})} < x_i < b_{i(x_{<i})}$, the finite-interval map is

$$x_i = a_{i(x_{<i})} + \left(b_{i(x_{<i})} - a_{i(x_{<i})}\right)t_i, \quad 0 < t_i < 1.$$

Its Jacobian is the product of the positive widths. Projective gauge changes, ray compactifications and endpoint charts supply their own Jacobians, each exactly once. On a negative causal cell the continuation is

$$(F - i0)^{q(\varepsilon)} = |F|^{q(\varepsilon)} e^{-i\pi q(\varepsilon)}.$$

The regulator-dependent phase must survive Laurent extraction. A sign-definite denominator does not make a complex numerator or amplitude positive.

The resulting boundaries can be algebraic. The native representation stores each exact defining polynomial and selected real branch, rather than demanding a radical formula. On a certified regular branch $P(x, r(x)) = 0$,

$$\partial_j r = -\frac{\partial_j P}{\partial_r P}.$$

Symbolica supplies symbolic derivatives and composed evaluator programs. Prepared native root evaluation supplies the selected section, and the map feeds the original numerator and smooth factors without expanding the full substitution. Colliding roots or $\partial_r P = 0$ require resolved coordinates; the regular implicit derivative cannot be extended through them by fiat.

2.7.1 Endpoint admission is a separate requirement

Integration is admitted only after the transported density has a supported endpoint form and its required closed-face behavior has been checked. Symbolic Taylor or IBP subtraction then produces integrable Laurent coefficients. The symbolic endpoint policy is independent of the contour-Jacobian choice. Generic open cells omit exceptional sets, but these cannot simply be discarded from a distributional causal integral: the regulated partition and its analytic continuation must retain their effect.

Current complete no-contour graph examples include the bubble, triangle, box, massless sunrise and massive sunrise in the overview. These establish useful sufficient classes, not the general algebraic resolver described in the research plan. Narrow single-chamber parameter templates also exist; they do not establish uniform support for arbitrary parametric families. The native CLI and HEPKit interfaces use the same admission boundary. Threshold support in WASM is outside this campaign.

Regular degree-six and degree-seven two-variable controls demonstrate that high degree alone is not the barrier: their causal polynomials are $y^6 - y + \frac{1+x}{16}$ and $y^7 + y - x$. Retained imaginary-part identities and cancelled pole coefficients check complete integrations, while higher-degree section tests check branch evaluation. Neither proves general collision resolution. Their supplementary measurements remain in `docs/no-contour-results.json`; they are not substituted into the eight-topology overview.

The difficult whole-atlas cases and the distinction between verified cells, admitted charts and executable sectors are developed in Chapter 5. The separate critical-locus discussion in Chapter 6 explains why an open-cell proof is also insufficient for certifying the absence of a pinch.

Benchmark findings and scientific checks

The opening tables support comparisons at a common finite Laurent order and approximately ten minutes, with the metric and clock qualifications in Chapter 1. The main lesson is that **a permitted contour displacement, a fast raw evaluator and a verified sign decomposition are each insufficient to predict the final uncertainty**. Sector proliferation, full Jacobians, importance sampling and precision rescue can dominate different cases. The following conclusions concern the recorded settings and independent confirmations, not universal rankings of the methods.

3.1 Scalar topologies: accuracy versus representation cost

The bubble, triangle, box, massless sunrise and massive sunrise have complete no-contour results with both samplers. Their no-contour kernel counts are 6, 18, 54, 6 and 62, respectively, versus one or two kernels in the corresponding contour recipes. The extra sign/endpoint charts can make a single kernel cheap while increasing the work of a full sweep. Fixed and dynamic contour kernels likewise have different arithmetic costs; the largest allowed deformation need not minimize either variance or runtime.

For the triangle and box, the dynamic construction improves QMC convergence at the recorded $S = 0.8$, $L = R = 1$ settings. The box's normalized radial relative SE is about 1.02×10^{-8} for both dynamic variants, compared with 5.68×10^{-5} for fixed strength and 5.33×10^{-10} for no-contour. Ordinary MC gives a different balance: its box values are about 8.5×10^{-5}, 1.39×10^{-4} and 9.73×10^{-4}, respectively. QMC and MC therefore cannot be ranked solely by raw call count. Polynomial and sign-aware bounds coincide for the relevant quadratic residual causal factors with no higher terms to distinguish their envelopes.

The massless sunrise is a non-crossing control, not evidence for treating an interior singularity. Its contour results nearly coincide after monomial extraction leaves no active deformation direction. The massive sunrise is a genuine above-threshold test: all seven verified cells enter through 62 closed endpoint charts in the no-contour representation. Complete generation takes 6.70 seconds, versus 1.44–1.61 seconds for the one-kernel contour recipes. At the recorded settings fixed deformation has the smallest reported SE for both samplers. The independently generated pySecDec Laurent reference is retained as a scientific control, rather than using this ranking as a test of correctness.

The all-massive kite favors fixed deformation over both dynamic baselines in this campaign. It has no complete no-contour result. The earlier three-massive/two-massless kite at small momentum is a separate tuning experiment: its results cannot be used as either a reference or a substitute for the physical $p^2 = 16$ row.

Roundoff is visible in the easiest cases. Bubble QMC replica SE reaches the binary64 rounding scale. Its value excludes common numerical bias and must not be read as accuracy beyond f64. The finite analytic value $2 - 3\frac{\log(3)}{2} + i\frac{\pi}{2}$ supplies an independent discrepancy check. Even a visually excellent SE or smooth QMC trend does not calibrate uncertainty coverage from one confirmation.

3.2 Precision rescue changes the effective cost

Long integration streams and deterministic f64 profiles exercise different points. In the triangle no-contour confirmations, QMC rescues 115,307,844 of 340,410,368 logical calls, while MC rescues all 61,915,136 calls. Both reach 256-bit precision with zero failures; the deterministic profile triggered no rescue. The corresponding massless-sunrise and box QMC fractions are 33.9% and 32.2%; their MC runs again rescue every logical call. Box maximum precision is 384 bits for QMC and 256 bits for MC, with zero failures. The recorded clocks include this work. These workloads have no per-trigger reason histogram, so rescue frequency alone does not identify its cause.

The massive-sunrise profile illustrates the same issue particularly clearly: its no-contour summed raw f64 cost is 47.4 microseconds per sweep, whereas the checked sweep costs approximately 23.7 milliseconds. Consequently, native root, branch and stability handling must be assessed together with the compiled expression. These observations motivate optimization but do not justify disabling mandatory precision rescue or dropping difficult points.

3.3 Double boxes: tuning restores fixed-like behavior

The scalar and physical D05 cases have 18 and 30 stochastic kernels. For the scalar input, complete fixed/polynomial/sign-aware generation takes 9.77/19.2/34.1 seconds. For the physical numerator the corresponding values are approximately 253/1800/1536 seconds, with six resident generation workers and measured peaks of 8.99/34.8/36.7 GB. These are complete publication measurements, not GCAD geometry timings. Tuned rows reuse the same artifacts and therefore repeat those generation costs.

The untuned $L = R = 1$ dynamic double-box results are substantially noisier than fixed $\lambda = 10^{-6}$, including large-uncertainty warnings. A separate calibration chooses $R = 10^6$ and small L as specified in Chapter 1. This weakens an overly restrictive displacement cap while controlling the strength and its spatial derivatives. No Jacobian or local-strength derivative is omitted. Sampling profiles are measured again at the chosen runtime settings; all eight final confirmation streams are independent of calibration.

Finite coefficient	Fixed	Dynamic tuned	Sign-aware tuned	Sampler
Scalar double box	1.02%	0.665%	0.755%	QMC
Scalar double box	0.246%	0.262%	0.233%	MC
Physical double box	1.98%	2.09%	2.14%	QMC
Physical double box	0.594%	0.662%	0.648%	MC

These radial relative SE values use the small correction to 600 seconds; actual durations are 600.11–604.79 seconds. Tuned dynamics improves the scalar QMC observation and recovers near-fixed behavior for the physical numerator. It does not establish a universal dynamic advantage. All eight confirmations have complete production coverage, no failed, unstable or discarded-zero points, and no norm-linearization warning.

Independent equal-work checks. Three scalar QMC controls each complete 71,958,528 accepted calls. The polynomial/fixed and sign-aware/fixed finite covariance-trace ratios are 0.193 and about 0.224. Their times are 145.95 and 136.80 seconds, against fixed's 112.49 seconds. Thus the scalar variance improvement is also present at equal work, using the full complex uncertainty rather than only its radial projection.

Three physical QMC controls each complete 179,896,320 calls. Fixed, polynomial and sign-aware take 377.81, 406.92 and 328.12 seconds, with covariance-trace ratios 1, 0.9904 and 0.9790. These single independent controls recover fixed-like per-call variance; they do not establish a statistically significant advantage. The estimates are not pooled with timed observations. Six MC work controls stop cleanly at their original 600-second caps and remain explicitly incomplete requested work. Their actual counts support only the declared heuristic work display.

For the polynomial construction, sufficiently small L approaches $\lambda = SL$ at fixed regular coordinates when the remaining bounds are inactive. Sign-aware smoothing leaves a small limiting correction retained in the native tests. Scalar test points with $L = 1.25 \times 10^{-6}$ and $R = 10^6$ give $\frac{\lambda}{SL} = 0.9792, 0.9915, 0.9985$; the larger scalar-QMC cap is not simply a copy of the fixed map. Such point diagnostics do not certify an arbitrary fixed contour globally or characterize every sector. In particular, the final weight summaries do not retain maximum-weight coordinates. The saved cap experiments discussed in the implementation chapter provide separate mechanism evidence, not new confirmation data.

3.4 Compatibility and independent references

Compatibility uses complete real–imaginary covariance for every available Laurent coefficient. For independent estimates the difference has covariance $\Sigma_1 + \Sigma_2$, and the descriptive distance is $\sqrt{\Delta^T (\Sigma_1 + \Sigma_2)^{-1} \Delta}$. Distinct reserved seeds support independence; no empirical cross-method covariance is invented. This estimated-covariance distance is not a calibrated p-value. Comparisons are correlated with one another, and a large collection of them is not a set of independent significance tests.

Baseline case	Estimates	Largest mutual distance, ε^0	Largest reference distance, ε^0
Massive bubble	18	2.39	2.39
Massive triangle	22	2.77	2.58
Massive box	22	2.46	2.17
Massless sunrise	22	3.24	3.08
Massive-threshold kite	18	3.25	—
Scalar D05 double box	12	2.72	—
Physical D05 double box	12	2.74	—

This baseline audit covers 126 retained native estimates from work/time arms. Its largest nonsingular distance across all orders is 3.779 between runs and 3.077 against a reference. The physical pole comparison with a noisy dynamic-QMC estimate produces the former. Singular and rounding-sensitive comparisons remain unresolved diagnostics; they are not regularized, padded or hidden inside these maxima.

The final double-box tuning extension independently checks 26 estimates over 234 complex Laurent blocks. Largest distances are 3.169 for scalar ε^0, 3.805 for physical ε^{-1} and 3.280 for physical ε^0, with no non-positive-covariance or rounding-sensitivity flags in that set. This extension does not discard the baseline's noisier observations or turn consistency into an analytic reference.

The massive-sunrise audit checks all eight fresh timed arms through orders $-2, -1, 0$. Its 84 complex-block comparisons have largest defined distance 2.611; five pole comparisons are rounding-sensitive and two covariance blocks are not positive definite. Those diagnostics are retained without regularization. Thirty-six component comparisons of the six contour arms with the independent pySecDec reference have maximum discrepancy 2.029 combined SE; the two no-contour arms have a separate 36-component audit against established controls with maximum 2.162. These are descriptive agreement checks, not calibrated coverage probabilities.

The bubble has the analytic finite value given above. Native HEPKit/OneLOop supplies triangle and box reference expressions at exactly the listed point, including their explicit inverse r_Γ measure factor. The massless sunrise uses Symbolica's Laurent expansion of the composed analytic bubble identity with its causal phase. The independently generated massive-sunrise pySecDec finite reference is $10.0161430322940 - 3.3089881368544i$. There is no independently established reference here for the five-massive kite at $p^2 = 16$ or either D05 double box. Their inter-method agreement is a consistency check; the scalar and physical integrals cannot validate each other by equality.

Native Numerica matrix operations perform the retained covariance checks. Reference rounding has no certified enclosure and is not silently converted into stochastic error. The compact evidence inventory in Chapter 1 preserves every association and qualification. The resulting campaign is reproducible without mistaking an unfinished atlas, a tiny sampling SE or a consistency check for a certified integral.

Implementation and performance

The methods in Chapter 2 share the same native source, symbolic subtraction and saved evaluator infrastructure. This chapter explains the engineering choices behind their different preparation and sampling costs. The focused measurements below isolate particular changes; they have their own settings and do not replace the complete-generation or ten-minute rows in the overview.

4.1 Preserve structure through generation and restoration

HEPKit and Linnet own graphs, models, routing and kinematics. Symbolica supplies factored expressions, differentiation, function maps, native duals, evaluator optimization and serialization. Numerica supplies arithmetic, matrices, RNG primitives and Havana. FastSecDec coordinates these native objects in Rust; the HEPKit bridge wraps them without a second algebra implementation.

Factored smooth densities and shared function bodies avoid repeatedly inserting large Jacobians into numerator terms. For one physical D05 source sector with 236 smooth terms, native structured construction and a shared determinant reduced the mapped record from 683.8 to 365.6 MB and sampled peak RSS from 2.178 to 1.288 GB. Mapping took 10.93 versus 7.95 seconds; a separate native debug compilation fell from 169.49 to 58.16 seconds. Saved full-vector parity passed at the admitted points, while saved program size increased by 6.6%. These are representation diagnostics for one sector, not full-generation speedups or repeated timing distributions.

Optimization settings. New generation defaults to ten Horner iterations and SymJIT O2. In the studied Symbolica revision, Horner zero takes the direct linearization route and skips subsequent Horner/common-instruction passes; it does not disable SymJIT O2. Earlier programs are never relabelled with the new default. Rebinding runtime strengths and caps restores the optimized program rather than repeating symbolic construction or Horner optimization.

For no-contour maps, the same structure-preserving principle keeps root sections and coordinate maps separate from the density they feed. Symbolic chain rules retain map and Jacobian derivatives. Finite endpoint series supply local coefficients, not approximations to bulk root functions. The unresolved whole-atlas problem in Chapter 5 is therefore distinct from whether a single high-degree root can be evaluated efficiently.

4.2 Symbolic subtraction with two Jacobian implementations

Both supported comparison arms keep endpoint reduction and subtraction symbolic. The symbolic Jacobian is differentiated and optimized with the complete density. The contour-only dual construction substitutes surviving first derivatives of the deformation image at final compilation using Symbolica's degree-one dual machinery. Higher endpoint derivatives have already been constructed symbolically. It does not use numerical-dual endpoint jets to reduce power singularities.

The full determinant and position-dependent strength are differentiated before this final representation choice. Coordinate seeds remain one on subtraction faces; parameter seeds remain zero. Exact-only boundary terms retain symbolic cancellation. Tests cover fixed and both dynamic prescriptions, higher endpoint poles, saved-program restoration and full-vector parity. Conditional bodies requiring hoisted image inputs remain unsupported by this opt-in compilation route. No change in the order of deformation and subtraction is assumed.

The following retained source-sector study is the matched evidence for this choice. Each pair uses the same 8,192 points, symbolic endpoints, SymJIT O2, **Horner zero** and common-pair cap 1000. Fixed uses $\lambda = 10^{-6}$; polynomial dynamics uses $S = .8$, $L = 10^{-6}$ and $R = 1$.

Contour / Jacobian	Time / peak RSS	Exact / JIT IR	Mean sample
Fixed / symbolic	559.9 s / 12.01 GB	108.3 / 156.2 kB	14.64 µs
Fixed / dual	566.9 s / 12.01 GB	136.8 / 202.8 kB	19.20 µs
Dynamic / symbolic	1511.6 s / 28.44 GB	294.0 / 524.6 kB	37.71 µs
Dynamic / dual	794.9 s / 16.73 GB	372.0 / 520.0 kB	29.99 µs

The timer includes generation, compilation, saving and pointwise checks; IR sizes exclude the complete owner. Both fixed arms rescue 27 points in double-double, both dynamic arms 28. Fresh-process restoration and complete Laurent-vector parity pass. This sector favors symbolic J for fixed mode and dual J for dynamic mode; dual J is not a universal optimization. Full evidence and the clarified mathematical ordering are retained in `docs/contour-deformation-results.json` and `docs/reviews/contour-symbolic-endpoints.md`.

4.3 Selective residency and native evaluator loading

Universal indexed artifacts allow a generation worker to persist one completed sector and exit. Serial integration loads only resident records; eviction releases the process before its replacement starts. The memory bound includes compact coordinator/source state and the active-sector peaks, including decoding, JIT and precision scratch. Whole-process measurements are necessary: dropping a Rust object alone does not prove that its pages returned to the OS. Ordinary discrete MC is the explicit residency exception in this campaign.

Records save Symbolica's optimized exact IR and native primary JIT payload. The latter restores the lowered application and callback descriptors; the owner still emits machine code. It is not a persisted machine-code cache. Exact IR remains available for eager, portable and precision-rescue paths. Immutable templates and buffers can be shared while mutable stacks and callback environments stay worker-local.

Two preserved records gave optimized cache-miss/cache-hit load times of 0.1804/0.1437 seconds (fixed) and 0.3865/0.3025 seconds (polynomial), with a 13.34% record-size increase. These historical loader probes do not measure current full-integral convergence. A separate 70.41 MB artifact ownership probe constructed 50 whole-owner clones: post-clone RSS fell from 6.68 to 1.79 GB and clone time from 3.32 to 0.78 seconds, while uncached loading stayed similar at 3.81 versus 3.71 seconds. It excludes binding, pilots and sampling.

Native owner corrections addressed callback-environment ownership (Symbolica PR 62) and remapping operands before common-subexpression lookup (PR 63). Their public consumer revision was `74225696`. These explain the measured implementation changes; old workspace test counts are not a current build certificate.

4.4 Deformation clearance and Jacobian amplification

The tuned double-box results in Chapter 3 show why caps are runtime parameters. A larger causal displacement is not necessarily a smoother integrand: map distortion and derivatives of the strength also affect weights. A supplementary mixed-mass kite study illustrates the mechanism. Its point $p^2 = \frac{3}{1000}$ with three massive and two massless lines is distinct from the five-massive $p^2 = 16$ overview case.

At one recorded large-weight point in the polynomial construction:

L	$\lambda(x)$	$\lvert F(z)\rvert$	$\lvert\det J\rvert$	**Weighted real output**
.1	.0800	1.087	1.040	.1722
1	.7989	1.088	5.322	−.7322
10	6.728	1.201	820.8	−108.94

The causal factor moves farther from zero, but the determinant amplifies the density. At $L = 10$ the full determinant is smaller than the frozen-strength diagnostic (821 versus 1194), so this observation does not justify dropping the strength derivative. Two independently confirmed seeds at tuned $L = 1$, $R = .25$, $S = .8$ reduced variance by 3.318 times relative to tuned fixed $\lambda = .4$; the polynomial evaluator's extra cost left mean variance-times-time at 1.044 times fixed. There was no demonstrated sampling-speed gain. `docs/contour-cap-scalar-results.json` retains the identities, replayed points, timings and full covariance. A determinant-distortion penalty is a possible future ansatz, not an implemented result.

4.5 Statistical ownership and focused reproduction

The caller owns scheduling and stream reservations. Pilots, production replicas and retries retain separate work identities; duplicate or stale returns cannot enter the estimate. Accepted complete QMC replicas or MC batches update the full vector covariance. Cancellation drains accepted work without turning an unfinished allocation or a missing sector into zero. Contour evaluation consumes no production RNG stream.

For quick generation studies, `generation.source_sectors = [2, 5]` selects original geometry-chart IDs before symmetry and subtraction. Artifacts retain the original total and local-to-original mapping and remain explicitly partial. Omit the option for a complete integral. The reproduction inventory in Chapter 1 associates each complete benchmark with its input, saved artifact, frozen executable and independent stream identities.

Support limits and algebraic boundaries

A verified generic-open decomposition is a geometry certificate, not a complete integration artifact. The source density must also admit consistent causal continuation, endpoint treatment and complete chart coverage. The supported scalar results meet their respective admission conditions; they do not establish a general resolver for arbitrary algebraic boundaries. The two difficult cases below make that distinction concrete.

The campaign stops at preparation for the no-contour double box. Neither the scalar nor the physical-numerator double box has a complete no-contour artifact or integral estimate. The massive kite also lacks a complete no-contour integral. Their successful contour comparisons remain in the opening tables. The measurements here describe **preparation diagnostics** and are not substitutes for missing generation, integration or convergence measurements.

5.1 Double box: complete geometry, incomplete density resolution

At the exact rational physical point, symGCAD solves and independently verifies **644,699 original generic-open cells** over the full positive projective domain, retaining both signs of F. The scalar and physical-numerator sources share this denominator geometry. Positive homogeneous reduction at x_0, conditional quadratic elimination in x_2, and lifting order $[x_0, x_1, x_3, x_4, x_5, x_6, x_2]$ produce the admitted decomposition. The internal monotone base has 464,265 cells; it is not a smaller complete original atlas.

Verified geometry measurement	Result
Native solver total	36.356 s
Solve process, including saved raw proof	156.526 s
Fresh verification process, including proof reload	142.072 s
Complete geometry trial wall time	298.601 s
Sampled aggregate peak RSS	11.875 GB
Saved raw native proof	5.284 GB

The native projection and lifting spans are 0.446 and 23.186 seconds; they are not an exclusive partition of the total. Serialization, decoding and independent verification account for much of the process time. Both children used one Rayon thread, under a 100 GB allowance. These are native geometry-driver observations, not a timed release-CLI generation of the regulated integral. The earlier parametric investigation did not complete; a general parametric double-box family is not established by the successful exact-point solve.

The remaining geometry is substantial. Later algebraic roots move with earlier coordinates, and 604,005 cells have an infinite boundary. Catalogue complexity in each lifting variable is:

Variable	Prefix nodes	Boundary polynomials	Max. degree
x_1	416	222	32
x_3	10,543	24	4
x_4	102,146	7	4
x_5	334,779	3	2
x_6	464,265	2	1
x_2	644,699	2	2

The degree is that of the boundary polynomial in the current lifting variable, not a specialized minimal polynomial. Root gaps, collisions, leading-coefficient zeros, infinity faces and intersections of closed faces require more than constant-root isolation or an interior sign sample.

5.2 What the local certificates establish

Native general improvements cover source-owned constant endpoints through degree 32, exact coefficient extraction for tight dyadic brackets, recursive fan replacements, original-certificate replay under positive rational scaling, and canonicalization of composed measures. Symbolica supplies the algebra; symGCAD retains selected-root and domain authority. Original denominators, ancestry, causal phases and each Jacobian remain explicit. These capabilities do not constitute an automatic resolution of every moving algebraic boundary.

For original double-box cell 0, the complete **156-term physical open density** was composed without expanding its algebraic roots inside the numerator. Four infinity charts passed 6,864 term comparisons, 44 summed-density checks and 154 independent coordinate/Jacobian checks at 128/256 bits. The density checks retain an independently evaluated lower-lip phase; directed-ball checks certify the tested coordinates and Jacobians. Exact finite-depth series also found a common c^3 infinity power in all 624 term/chart combinations, with epsilon cancelling. Those local coefficients are not truncated bulk integrands or certificates for the remaining direction faces and causal seams.

A separate scalar admission diagnostic completed **307 closed charts of one original cell**, retaining the full regulator density on the common sufficient strip $-\frac{3}{2} < \Re(\varepsilon) < 1$. It took 4,122.9 seconds and peaked at 9.96 GB aggregate RSS in a debug preparation driver. It produced **zero compiled kernels and zero integral estimates**. It does not admit the physical numerator. Its source-bound objects ended with the process; saved receipts alone do not reconstruct their source authority.

Catalogue sharing reduces 3,868,194 per-cell axis occurrences to 1,556,848 prefix nodes, a factor of 2.485; repeated root-bound uses reduce by 3.003. The 260 distinct global boundary polynomials and 45 global support classes do not prove equality of branches, maps, densities or closed domains. Compatibility merging retains original cells and excluded seams; no measured merged endpoint atlas is available.

For scale only, one chart per original cell would give 644,699 charts; 307 per cell would give 197,922,593. These are **illustrations, neither predictions nor bounds**. One admitted cell does not support a whole-atlas sector-count, runtime, memory or artifact-size forecast. The unresolved closed-family continuation/compiler bridge remains separate from the successful geometry.

5.3 Massive kite: a boundary is not a complete interval

The equal-five-mass kite at $p^2 = 16$ has **9,840 verified original cells**. A selected fourth boundary closes over three integration directions in all **ten replacement charts of cell 0**, passing **40 strict checks at 128/256 bits**. That result does not close the fourth interval. Its width, denominators and root-gap intersection with the upper face must join the full original U, F, numerator and Jacobian inventory. In particular, $F^{-1-2\varepsilon}$ requires common-strip continuation and symbolic subtraction.

A native divided-difference probe verifies candidate width and root identities, including higher-degree controls. Positivity on every closed face, complete interval densities, automatic coverage of the other 9,839 cells and full Laurent compilation remain unfinished. This is a substantive implementation gap, not evidence that the integral is physically inadmissible or that GCAD failed. A future general resolver must cover these obligations before claiming whole-integral support.

The immutable recipes, measurements and source/reuse audits are indexed in `docs/reviews/no-contour-final-scope.md`, `docs/no-contour-gghh-verified.json` and the `examples/no_deformation/gghh_feasibility/` inputs. Historical failed trials and local diagnostics remain available there; they are not benchmark rows.

Critical loci and future region analysis

The constructions in Chapter 2 handle regular threshold sheets and admitted endpoint charts; Chapter 5 records their unresolved boundaries. Here we examine exceptional critical loci, exact scratch evidence and prospective local resolution and region analysis. These proposals are not implemented capabilities or new benchmarks.

6.1 Stationary zeros, causal pinches and present diagnostics

For the weighted-gradient map, $v_i = x_i(1-x_i)\partial_i F$ and $z = x - i\lambda(x)v$. At an interior stationary zero $F(x_*) = 0$, $\operatorname{grad} F(x_*) = 0$, every finite smooth strength leaves $z(x_*) = x_*$. Increasing the safety factor or caps cannot displace it. The Jacobian need not be one: derivatives of v survive even when $v = 0$. A genuine causal pinch cannot be removed by an admitted nonsingular contour homotopy. This is a limitation of gradient deformation already recognized in Borowka, Carter and Heinrich.

A denominator zero, a trapped contour, singular kinematic dependence and a pole in dimensional regularisation are different statements. Numerators and the physical sheet affect the answer; a pinch may make a derivative singular without making the integral divergent (Collins, §II.1.4). Here the homogeneous conditions $F = \operatorname{grad} F = 0$ with $U \neq 0$ concern finite-momentum, first-type candidates. Second-type singularities involve infinity and require additional strata. Critical equations for the Lee–Pomeransky polynomial $G = U + F$ have a different interpretation; see Fevola, Mizera and Telen. The chart gradient uses independent coordinates; faces, vanishing U and singular coordinate maps cannot be classified by the interior test alone.

Current detection is encounter-dependent, not an exhaustive search. After certain terminal nonfinite or precision failures, FastSecDec's cold diagnostic checks retained F and $A = \sum_i x_i(1-x_i)(\partial_i F)^2$ with native certified balls. It reports a stationary-zero context only when both enclosures are exactly zero, not merely when they contain zero. This remains available with optional validation off and adds no hidden per-sample checking. A successfully evaluated, natively cancelled density is allowed; the diagnostic does not identify a unique failing branch. In the strict interior, $A = 0$ means $\operatorname{grad} F = 0$; on a face, zero weights can hide normal derivatives.

A finite pilot can miss a lower-dimensional locus. Neither large weights nor small strengths certify its presence. For example, $F_0 = \left(x - \frac{1}{2}\right)^2$ and $F_1 = 1 + \left(x - \frac{1}{2}\right)^2$ have identical derivative envelopes for constant positive U. At $x = \frac{1}{2}$, the polynomial prescription gives $\lambda = SL$ in both cases: only their displacement vanishes. A complete preflight would instead return **certified absent**, **certified loci found**, or **unresolved**, with evidence renewed on rebinding unless a chamber proof applies.

6.2 Equality strata: how exact algebra and CAD can cooperate

Full CAD can represent lower-dimensional solution cells. The current symGCAD generic-open interface does not return them as cells. This matches Vitaly's original implementation, which accepts strict inequalities and omits the algorithm's separate boundary list; it is not a porting regression. Retained exceptional polynomials locate possible missing strata, but do not resolve those strata.

The native control $0 < x < 1$, $F = (2x-1)^2$ returns two verified open intervals, with no critical zero-dimensional cell, although exact substitution gives $F\left(\frac{1}{2}\right) = F'\left(\frac{1}{2}\right) = 0$. Equality input is rejected. The original implementation returns the same intervals for $F > 0$; its optional merge absorbs the zero seam. Neither a completed generic solve nor equal signs across a seam prove absence of a critical locus. Integration can nevertheless use open cells if their partition identity is established in a common regulated convergence domain and continued correctly. For example,

$$\int_{-1}^{1} (x - i0)^{-1-\varepsilon}\,\mathrm{d}x = \frac{1 - e^{i\pi\varepsilon}}{-\varepsilon} \longrightarrow i\pi.$$

Initially $\operatorname{Re}\varepsilon < 0$; the continued phases preserve the boundary contribution without a separate point integral. Measure zero alone would not justify discarding its singular neighbourhood.

Native Symbolica derivatives, Groebner bases and algebraic solutions reduce critical equations. On the positive projective interior, adjoining $h\prod_i x_i - 1 = 0$ excludes coordinate boundaries. A unit ideal proves absence; a non-unit ideal needs real-domain tests. An exact parametrization requires original-equation substitution and the converse under nonzero guards; symGCAD verifies the remaining inequalities. Otherwise, complete equality/section solving or recursion on exceptional strata is needed. A native nonvanishing certificate for $F^2 + \sum_i (\partial_i F)^2$ also proves absence; failure to obtain one is inconclusive.

6.2.1 Exact controls at the benchmark points

The scratch calculations use native graph polynomials, exact kinematics and the **strict positive projective interior**, fixing the last parameter to one. They do not classify boundary loci or entire kinematic chambers.

Case	Exact point	Certified interior outcome
Massive bubble	$m^2 = 3, s = 16$	Absent: saturated ideal (1)
Massive triangle	$m^2 = 1, s = 5$	Absent: saturated ideal (1)
Massive box	$m^2 = 1, s = 5, t = -1$	Absent: saturated ideal (1)
Massless / massive sunrise	$s = 1$ / $m^2 = 1, s = 16$	Absent: saturated ideals (1)
Massive kite	$m^2 = 1, s = 16$	Absent: strictly positive ideal member
Bubble / sunrise thresholds	$m^2 = 3, s = 12$ / $m^2 = 1, s = 9$	Found at all parameters one; $U = 2$ / 3
Three-loop crown	$s = 1, t = -\frac{1}{5}$	Found: four-dimensional projective family

The triangle has two massless external legs; the box has massless external legs and $u = -4$. All five kite masses equal one. For the kite, an independently checked combination of its five homogeneous derivatives equals $1 + x_0 + x_1 + x_2 + x_3 + 16x_0x_3 > 0$. Its complex critical variety need not be empty. These exact exclusions are small algebra diagnostics, not integral-generation timings. The threshold controls show that critical points are not confined to three loops. No double-box critical-locus classification was performed.

6.2.2 The crown: retain tangent and normal directions separately

The three-loop, eight-propagator massless crown polynomial comes from Gardi, Herzog, Jones and Ma, §3.3, also treated by Olsson and Stone. With zero-based labels, set $x_1 = 1$ and use

$$a = x_0, \quad b = x_3, \quad c = x_5, \quad d = x_7, \quad x_2 = ab + u, \quad x_4 = ac + v, \quad x_6 = ad + w.$$

Native substitution and matrix operations verify

$$F = (s + t)duv - tcuw - sbvw, \quad \det H_\perp = 2st(s + t)bcd.$$

For $st(s + t) \neq 0$ and positive b, c, d, the normal equations force $u = v = w = 0$. All eight original derivatives and F then vanish, proving conditional completeness of this positive interior family. It has four free positive projective parameters and $U|_{u=v=w=0} = (a + 1)^3(bc + bd + cd + bcd) > 0$. GCAD verifies these reduced inequalities as one positive-U cell; it does not discover the locus by itself. The full projective Hessian has rank three, so an isolated-point treatment would be inappropriate.

The scratch process took 0.0432 s and 12.7 MiB, using warm native libraries; reduced GCAD solve/verification took 7.36/2.31 ms. No crown integrand was generated or integrated. Exact revisions, commands and independent review are in `docs/reviews/gcad-critical-loci.md` and `docs/gcad-critical-loci-results.json`; these observations do not time a general critical-locus solver.

6.3 From a certified pinch to local subtraction

Local power counting determines whether a pinch produces a divergent value, a threshold branch or a regulator pole. For an isolated definite quadratic point with regular nonzero numerator and U, the model is $F = \tau + |y|^2 +$ higher orders. For $\operatorname{Re}\tau > 0$, $\operatorname{Re}\beta > \frac{m}{2}$,

$$\int_{\mathbb{R}^m} \frac{\mathrm{d}^m y}{(\tau + |y|^2)^\beta} = \pi^{\frac{m}{2}} \frac{\Gamma(\beta - \frac{m}{2})}{\Gamma(\beta)} \tau^{\frac{m}{2} - \beta}.$$

With a local cutoff, continuation determines the leading nonanalytic term up to analytic terms and smooth factors. Negative powers can diverge; positive noninteger powers can remain finite; integer limits can produce logarithms if their coefficients survive. Numerator zeros change the power. Apparent Gamma poles can cancel with the cutoff's analytic part. At $\tau = 0$ with $\beta(\varepsilon) = \frac{m}{2} + b\varepsilon$, the marginal radial model gives

$$\int_0^{r_0} r^{-1-2b\varepsilon} \, \mathrm{d}r = \frac{r_0^{-2b\varepsilon}}{-2b\varepsilon}, \quad \operatorname{Re}(b\varepsilon) < 0.$$

This exposes a **possible** dimensional pole, whose residue must survive the complete sum. It does not establish a universal epsilon pole at every pinch. Degenerate or indefinite forms and critical manifolds require their own normal/tangent analysis, not substitution of Hessian rank for m.

6.3.1 A targeted hybrid can avoid the full sign atlas

One prospective route resolves only certified critical neighbourhoods and retains the contour elsewhere. It need not partition every regular point of $F = 0$. Pinch-dissection precedents exist in Gardi et al. the following local argument is a separate, unimplemented construction.

At an isolated stationary point with nonsingular real Hessian H, $W_* = \mathrm{diag}(x_i(1 - x_i)) > 0$, $U \neq 0$ and finite $\lambda_* > 0$, write $x = x_* + y$. The same smooth global contour gives

$$z = x_* + My + O(|y|^2), \quad M = I - i\lambda_* W_* H,$$

$$\mathrm{Im}\, F(z) = -\lambda_* y^T H W_* H y + O(|y|^3).$$

The leading imaginary form is strictly negative for nonzero real y, even for an indefinite H, and M is invertible. The analytic expansion uses ordinary transpose, not conjugate transpose. Local-strength derivatives remain in the full Jacobian and affect higher terms.

Split a certified cube $|y_i| \leq \delta$ into orthants and largest-coordinate wedges. In each, use $y_p = \delta s_p r$, $y_j = \delta s_j r t_j$ for $j \neq p$, with $0 \leq r, t_j \leq 1$ and $s_j \in \{-1, 1\}$. The real volume factor is $\delta^d r^{d-1}$ and exact stationarity extracts $F(z(x_* + \delta r a(t))) = r^2 Q(r, t)$. The value at $r = 0$ is the smooth derivative-defined extension, never numerical $\frac{0}{0}$. A certified higher-order remainder yields $\mathrm{Im}\, Q \leftarrow c < 0$ on the closed angular charts for a sufficiently small cube. The radial density then has the form $r^{d-1-2\beta(\varepsilon)} h(r, t; \varepsilon)$ and admits symbolic subtraction.

This favorable construction starts with at most $d2^d$ local charts per isolated point; it excludes complement charts and later kernel splitting. For the crown, a transverse analogue would retain four tangent parameters and start locally with $3 \times 2^3 = 24$ normal charts, subject to uniform bounds. This is neither a global sector forecast nor the paper's different 24-ordering dissection. The double-box's 644,699 sign cells count neither critical points nor local pinch charts.

Preserving one global contour across artificial interfaces is essential: resetting face weights on each new cube can freeze regular threshold zeros on the seams. Exact centres, full numerators/factors/Jacobians, coverage and all closed-face derivatives must be retained. Degeneracies, tangent-boundary collisions and $U = 0$ require further resolution. Subtraction identities must first hold in a common regulated family; additional analytic regulators can be necessary (pySecDec, §4.8). No numerical failure or unresolved chart becomes a zero contribution.

6.4 Prospects for complete expansion by regions

The natural objective is a **complete, possibly redundant asymptotic expansion to a requested order**, with coverage and remainder control. A minimal list of nonzero regions is a harder separate problem. Exact critical-locus analysis could improve construction, but it does not supply this theorem by itself.

6.4.1 Position relative to the literature

Pak–Smirnov geometry associates scalings with Newton-polytope faces. The asy2 method already exposes potential and Glauber regions in its examples; pySecDec's regions implementation discusses cancellations in Minkowski kinematics. A hidden region can become a facet region after a change of variables: "interior" refers here to parameter space, not an invariant location in the original exponent polytope.

Gardi's Hidden Region Finder (HRF), September 2026 is the closest comparator. It connects cancellation loci, dissection and asymptotic scaling; §2.4.3 proposes a scaling-first face/positive-Landau search as a theoretical alternative to its negative audits. Its study treats massless, unit-numerator examples and leaves massive and numerator-dependent extensions open. SOFIA, by Correia, Giroux and Mizera, instead studies candidate kinematic singularities and alphabets via Baikov representations and Fubini reduction. It is not a facet-only regions tool. Certified local exponents and physical-sheet analysis could complement its outputs.

6.4.2 What a completeness claim would require

The limiting family carries information absent from its critical set: $F_2 = (x-a)^2 + \rho$ and $F_4 = (x-a)^4 + \rho$ share the same limiting stationary zero but have widths $\rho^{\frac{1}{2}}$ and $\rho^{\frac{1}{4}}$. Neither has a real stationary zero at positive ρ. A solve at one finite kinematic point is insufficient. A viable certificate contract would require:

1. An exact limiting family, physical chamber, causal sheet and target order, including projective faces, infinity and relevant $U = 0$ intersections.
2. Certified real solutions of the relevant **leading** systems. Scaling can promote cancellations absent from the stationary set of the unscaled F.
3. Local charts retaining multiplicities, tangent parameters, branches and Jacobians; resolution of nested or colliding strata and the regular complement; ordinary boundary regions alongside hidden ones.
4. A partition or explicit overlap prescription, and bounds on the integrated remainder in a common convergence domain before analytic continuation. Factors $\rho^{a+b\varepsilon}$ must survive until the Laurent expansion.

Equality-aware CAD and variety sub-CAD can avoid irrelevant ambient cells under their respective hypotheses (England, Bradford and Davenport and Wilson et al.). Degenerate equations and vanishing projection polynomials need a complete fallback. The current symGCAD open-cell interface is not that equality solver. Jantzen's regional identities require covering/convergence assumptions and overlap terms. Regulator-pole cancellation cannot certify completeness: an omitted contribution can be finite. Leading-order vanishing does not justify dropping all higher orders; exact coefficient cancellation may require integral identities.

6.4.3 Where symGCAD and the native implementation could help

Opportunity	Actual foundation	Prospective gain / missing work
Target critical geometry	Scalar exclusions; crown's exact normal reduction.	Avoid a global sign atlas; general component coverage still needed.
Reject inadmissible candidates	Exact positivity/nonvanishing certificates and kite identity.	Reduce expensive real-domain work; failed certificates remain unresolved.
Reduce and reuse algebra	Linear/quadratic reductions, positive-root covers, bounded ordering, exact caches.	Share guarded geometry across faces/families; preserve exceptional strata.
Treat the full density	Native numerators, symbolic derivatives, composed root maps and subtraction.	Massive algebraic branches and numerator-dependent orders; general family resolution and remainder proofs missing.

The inventory is pinned to symGCAD `36bfd9e` checkpoint; some options remain experimental. Its matched study reports a 6.82-fold geometric-mean improvement in generic-CAD algebra phases over the original implementation. This is not an end-to-end regions speedup, and unresolved cases remain. The most concrete extension beyond HRF's stated scope is massive families with algebraic branches and realistic numerators, not an already demonstrated faster or complete region finder.

A first comparative study should reproduce Euclidean regions, massive threshold controls, the crown and HRF's moving/boundary examples. Compare exact scalings and coefficient expansions before speed; record discovery, resolution, verification, chart counts and memory, including negative cases and incomplete searches. The acceptance criterion is certified coverage with a controlled remainder. Literature and reuse evidence are retained in `docs/reviews/region-analysis-prospects.md`; pinch assumptions and source audits are in `docs/reviews/landau-pinches.md`.